

Program : Diploma in Renewable Energy	
Course Code: 5291	Course Title: Induction & Synchronous Machines
Semester : 5	Credits: 4
Course Category: Program core course	
Periods per week: 4 (L:3 T:1 P:0)	Periods per semester: 60

Course Objectives:

- To comprehend the working of synchronous machines.
- To comprehend the working of induction machines.

Course Prerequisites:

Topic	Course Code	Course name	Semester
Electromagnetic Induction	2321	Basic Electrical Engineering and Renewable Technology	2
DC Machines and transformers	4292	DC Machines and transformers	4

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive Level
CO1	Identify the power stages and equivalent circuit of three phase induction motor.	15	Applying
CO2	Identify the performance of three phase induction motors and working of single phase motors	15	Applying
CO3	Develop the EMF equation and vector diagrams of synchronous generator	13	Applying
CO4	Identify the performance of the synchronous generator and motor.	15	Applying
	Series Test	2	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3						
CO2	3						
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Identify the power stages and equivalent circuit of three phase induction motor.		
M1.01	Identify the construction and working of three phase induction motors	4	Applying
M1.02	Explain various torques in three phase induction motor	3	Understanding
M1.03	Illustrate the power stages of a three phase induction motor and solve problems	4	Applying
M1.04	Develop the equivalent circuit of a three phase induction motor	4	Applying

Contents:

Three phase induction motor-production of rotating magnetic field- construction - slip ring and squirrel cage rotors-synchronous speed-motor speed- Slip- frequency of rotor induced EMF - rotor power factor - simple problems.

Torque in an induction motor- relation between torque and rotor power factor- concept of standstill - derivation of condition for maximum starting torque- -basic concept and equations of torque under running conditions , maximum torque and full-load torque

Power Stages :Losses and efficiency- power stages- problems-concept of torque in synchronous watts-concept of crawling and cogging- Applications of induction motors

Equivalent Circuit of induction motor: Induction motor as a generalized transformer – equivalent circuits- simple problems- phasor diagrams.

CO2	Identify the performance of three phase induction motors and working of single phase motors		
M2.01	Choose various tests to pre-determine the performance of three phase induction motors	5	Applying
M2.02	Illustrate various starting and speed control methods for three phase induction motors.	3	Understanding
M2.03	Explain the construction and equivalent circuit of double cage induction motor	2	Understanding
M2.04	Illustrate the construction and working of various single phase motors	5	Understanding
	Series Test – I	1	

Contents :

Testing of three phase induction motors - no load and blocked rotor tests - circle diagram – pre determination of performance and maximum quantities - problems.

Starting and speed control of induction motors - various starters - basic working and diagrams-Speed control of induction motors - different methods-Braking of induction motors - different methods

Double cage induction motor - construction - equivalent circuit- advantages- applications

Single phase motors -Double field revolving theory - working with basic diagrams of split phase , capacitor start induction run motor , capacitor start capacitor run motor , permanent capacitor motor , shaded pole motor, AC series motor, universal motor, repulsion motor-Applications

CO3	Develop the EMF equation and vector diagrams of synchronous generator		
M3.01	Illustrate the construction and working of synchronous generators.	3	Understanding
M3.02	Develop the emf equation of a synchronous generator.	4	Applying
M3.03	Explain the effect of power factors on armature reaction	3	Understanding
M3.04	Illustrate the vector diagrams of a loaded alternator	3	Understanding
Contents : Synchronous generators-working principle -production of sinusoidal alternating emf-constructional features - stator - advantages of having stationary armature - types of armature windings(Listing only)- rotor-classifications -salient type-cylindrical type - comparison between salient and cylindrical rotors -damper winding Generated EMF -Relation between rotor speed and frequency of generated emf-pitch factor and distribution factor (definitions only) -derivation of emf equation-problems Alternator on load - armature resistance - leakage reactance - armature reaction - Effect of power factors (upf, zpf lag and zpf lead) on armature reaction with diagrams Vector diagrams of a loaded alternator-synchronous reactance -Synchronous impedance- phasor diagrams for various power factors(diagrams only)			
CO4	Identify the performance of the synchronous generator and motor.		
M4.01	Select various tests to pre-determine the voltage regulation of synchronous generators.	6	Applying
M4.02	Explain parallel operation of alternators.	2	Understanding
M4.03	Summarize the operation of synchronous motors	3	Understand
M4.04	Illustrate the V and inverted V curves of synchronous motor	4	Understand
	Series Test – II	1	

Contents :

Tests on alternator -Regulation of alternators -definition - testing of alternator -open circuit test - short circuit test -calculation of synchronous impedance and reactance
Predetermination of voltage regulation - methods - EMF method – MMF method – ZPF method - Problems

Parallel operation of alternators -need -conditions - Synchronization of three phase alternators-methods-basic concepts of dark lamp , bright lamp and synchroscope - Synchronizing current - synchronizing power - synchronizing torque (definitions and equations only)

Synchronous motor - construction -working principle-Effect of load on synchronous motor - changing load with constant excitation - changing excitation with constant load (phasor diagram approach only) -Power developed by synchronous motor(equation only).

Losses and starting of synchronous motor - Losses and power flow diagram - V and inverted V curves -methods of starting -procedure for starting

Text / Reference

T/R	Book Title/Author
T1	BL Theraja. Electrical technology. Vol- II: S Chand & co.
T2	J.B. Gupta. Theory & Performance of Electrical Machines: S. K. Kataria & Sons
T3	Special Electrical Machines:E G Janardanan
R1	Electrical Machinery : Dr. P.S. Bimbhra

Online Resources

Sl.No	Website Link
1	https://nptel.ac.in/courses/108106072
2	NPTEL courses: https://nptel.ac.in/courses/108/102/108102146
3	https://nptel.ac.in/courses/108/106/108106071